$$P_2(z) = az^2 + bz + c \geqslant 0 \qquad \forall z \in \mathbb{R}$$

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4 \cdot a \cdot c}}{2a}$$

$$\underline{b^2 - 4ac < 0}$$

$$Var(aX+bY) = a^2 Var(X) + 2ab\,Cov(X,Y) + b^2 Var(Y)$$

$a = z$

$b = 1$

$$Var(zX+Y) = \underbrace{z^2 Var(X) + 2z\,Cov(X,Y) + Var(Y)}_{P_2(z)} \geq 0$$

$$[2\,Cov(X,Y)]^2 - 4\,Var(X)\,Var(Y) \leq 0$$

$$(Cov(X,Y))^2 \leq Var(X)\,Var(Y)$$

$$\left[\frac{Cov(X,Y)}{\sigma_X \sigma_Y}\right]^2 \leq 1$$

$$\rho_{X,Y}^2 \leq 1$$

$$\rho_{X,Y} \in [-1, +1]$$